MATHEMATICAL GAME THEORY

Final Examination: Solutions and Rubric

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Grading philosophy. The rubric lists the essential steps of each argument. Alternative correct proofs receive full credit. Deduct for gaps only where a step that needs justification is asserted without it. Numerical answers must be exact.

Problem 1 *Correlated equilibrium* (16 points)

(a) Pure: (D,C) with payoffs $(8,2)$ and (C,D) with $(2,8)$; (D,D) and (C,C) are not equilibria ($2>0$ and $8>6$ respectively). Mixed: if the opponent dares with probability q, $u(\mathsf{D}) = 8(1-q)$ and $u(\mathsf{C}) = 2q + 6(1-q) = 6 - 4q$, which are equal at $q = \frac{1}{2}$. So each player dares with probability $\frac{1}{2}$ and earns 4.

RUBRIC. 1 pt: pure equilibria. 1 pt: mixed probabilities. 1 pt: payoffs.

(b) For the row player, the recommendation D occurs with weights $(z_{\mathsf{DD}}, z_{\mathsf{DC}})$ on the column's action. Obeying D rather than switching to C requires

$$0 \cdot z_{\mathsf{DD}} + 8 z_{\mathsf{DC}} \;\geq\; 2 z_{\mathsf{DD}} + 6 z_{\mathsf{DC}} \iff z_{\mathsf{DC}} \geq z_{\mathsf{DD}}.$$

Obeying C (weights $z_{\mathsf{CD}}, z_{\mathsf{CC}}$) requires $2z_{\mathsf{CD}} + 6z_{\mathsf{CC}} \geq 0 \cdot z_{\mathsf{CD}} + 8 z_{\mathsf{CC}} \iff z_{\mathsf{CD}} \geq z_{\mathsf{CC}}$. By symmetry the column player's constraints are $z_{\mathsf{CD}} \geq z_{\mathsf{DD}}$ and $z_{\mathsf{DC}} \geq z_{\mathsf{CC}}$. Together with $z \geq 0$ and $\sum z = 1$:

$$\boxed{z_{\mathsf{DC}} \geq z_{\mathsf{DD}}, \quad z_{\mathsf{CD}} \geq z_{\mathsf{CC}}, \quad z_{\mathsf{CD}} \geq z_{\mathsf{DD}}, \quad z_{\mathsf{DC}} \geq z_{\mathsf{CC}}}.$$

(Unconditional form: multiplying a conditional constraint by the probability of the recommendation does not change it, so there is no need to divide by that probability.)

RUBRIC. 2 pts: correct form of one obedience constraint. 3 pts: all four reduced inequalities. Writing constraints only for recommendations of positive probability is acceptable.

(c) The total payoff is

$$W = 0 \cdot z_{\mathsf{DD}} + 10 z_{\mathsf{DC}} + 10 z_{\mathsf{CD}} + 12 z_{\mathsf{CC}} = 10(1 - z_{\mathsf{DD}}) + 2 z_{\mathsf{CC}} \leq 10 + 2 z_{\mathsf{CC}}.$$

From $z_{\mathsf{CC}} \leq z_{\mathsf{DC}}$ and $z_{\mathsf{CC}} \leq z_{\mathsf{CD}}$, we get $3 z_{\mathsf{CC}} \leq z_{\mathsf{CC}} + z_{\mathsf{DC}} + z_{\mathsf{CD}} \leq 1$, so $z_{\mathsf{CC}} \leq \frac{1}{3}$ and $W \leq \frac{32}{3}$. Equality requires $z_{\mathsf{DD}} = 0$ and $z_{\mathsf{CC}} = \frac{1}{3}$, and then $z_{\mathsf{DC}} = z_{\mathsf{CD}} = \frac{1}{3}$ (both are at least $\frac{1}{3}$ and they sum to $\frac{2}{3}$). This distribution satisfies all four constraints. Hence the unique welfare-maximizing CE is

$$\boxed{z = (0,\ \tfrac{1}{3},\ \tfrac{1}{3},\ \tfrac{1}{3})}, \qquad W = \tfrac{32}{3}, \qquad \text{each player earns } \tfrac{1}{3}(8+2+6) = \tfrac{16}{3}.$$

Each player gets $\frac{16}{3} > 4$ (the mixed NE), and the total $\frac{32}{3}$ exceeds the total 10 of either pure NE, while treating the players symmetrically.

RUBRIC. 1 pt: expression for W. 2 pts: upper bound $z_{CC} \le \frac{1}{3}$ from the constraints. 1 pt: optimum and uniqueness. 1 pt: comparison. An answer found by a correct LP set-up with no optimality proof: 3/5.

(d) Let (σ_1, σ_2) be a Nash equilibrium and $z = \sigma_1 \times \sigma_2$. For player 1, recommendation s_1, and deviation s_1',

$$\sum_{s_2} z(s_1, s_2)\big[u_1(s_1, s_2) - u_1(s_1', s_2)\big] = \sigma_1(s_1)\big[u_1(s_1, \sigma_2) - u_1(s_1', \sigma_2)\big].$$

If $\sigma_1(s_1) = 0$ this is 0. If $\sigma_1(s_1) > 0$, then s_1 is in the support of an equilibrium strategy and hence is a best response to σ_2, so the bracket is ≥ 0. The same argument applies to player 2. Hence z is a CE. □

RUBRIC. 2 pts: factorization using independence. 1 pt: support strategies are best responses.

Problem 2 *Shapley value and the core* (18 points)

(a) Fix an ordering $\pi = (i_1, \dots, i_n)$. The marginal contributions telescope:

$$\sum_{k=1}^{n} \Big[v(\{i_1, \dots, i_k\}) - v(\{i_1, \dots, i_{k-1}\}) \Big] = v(N) - v(\varnothing) = v(N).$$

Averaging over the $n!$ orderings, $\sum_i \phi_i(v) = \frac{1}{n!} \sum_\pi v(N) = v(N)$. If i is a null player, every marginal contribution $v(P_i^\pi \cup \{i\}) - v(P_i^\pi)$ is 0 (take $S = P_i^\pi$), so $\phi_i(v) = 0$. □

RUBRIC. 3 pts: telescoping sum for a fixed order and averaging. 2 pts: null player.

(b) Marginal contributions $(1, 2, 3)$ by ordering:

Order	Player 1	Player 2	Player 3
1, 2, 3	0	8	2
1, 3, 2	0	4	6
2, 1, 3	8	0	2
2, 3, 1	6	0	4
3, 1, 2	6	4	0
3, 2, 1	6	4	0
Sum	26	20	14

$\phi = \boxed{\left(\frac{13}{3}, \frac{10}{3}, \frac{7}{3}\right)}$, which sums to 10.

RUBRIC. 3 pts: correct marginal contributions (or the equivalent subset formula). 2 pts: final values. One arithmetic slip in the table: -1.

(c) The core is $\{x : x_1 + x_2 + x_3 = 10,\ x_i \ge 0,\ x_1 + x_2 \ge 8,\ x_1 + x_3 \ge 6,\ x_2 + x_3 \ge 4\}$. Using $x_1 + x_2 + x_3 = 10$, the pair constraints become $x_3 \le 2$, $x_2 \le 4$, $x_1 \le 6$; these imply $x_i \ge 0$ (e.g.

$x_1 = 10 - x_2 - x_3 \geq 4$). So the core is the triangle

$$\{x : x_1 + x_2 + x_3 = 10,\ x_1 \leq 6,\ x_2 \leq 4,\ x_3 \leq 2\}$$

with extreme points $\boxed{(6,4,0),\ (6,2,2),\ (4,4,2)}$; it is nonempty. The Shapley value has $x_3 = \frac{7}{3} > 2$, i.e. $x_1 + x_2 = \frac{23}{3} < 8 = v(\{1,2\})$: coalition $\{1,2\}$ blocks, so the Shapley value is **not** in the core.

RUBRIC. 2 pts: core inequalities rewritten as upper bounds. 2 pts: extreme points. 1 pt: Shapley value not in core, naming the blocking coalition.

(d) Player A (weight 3) is pivotal exactly when A is not first, since any single other player plus A reaches 5 but B and C together have only 4: orders BAC, BCA, CAB, CBA. B is pivotal only in ABC and C only in ACB. Index: $\boxed{(\frac{2}{3},\ \frac{1}{6},\ \frac{1}{6})}$. ($A$ holds $\frac{3}{7}$ of the votes but $\frac{2}{3}$ of the power.)

RUBRIC. 2 pts: pivot counts. 1 pt: index.

Problem 3 *Evolutionary stability* (18 points)

(a) If (σ,σ) is a strict NE, condition (i) holds for every $\tau \neq \sigma$, so σ is an ESS. Conversely, if σ is an ESS, then for every $\tau \neq \sigma$ either (i) or the equality in (ii) holds; in both cases $u(\sigma,\sigma) \geq u(\tau,\sigma)$, so σ is a best response to itself and (σ,σ) is a Nash equilibrium. □

RUBRIC. 2 pts each direction.

(b) *Pure strategies.* $u(D,H) = 0 > -2 = u(H,H)$, so H is not a best response to itself and is not an ESS. $u(H,D) = 6 > 3 = u(D,D)$, so D is not an ESS.

Mixed equilibrium. Against $q = \Pr(H)$: $u(H,q) = -2q + 6(1-q) = 6 - 8q$ and $u(D,q) = 3(1-q)$. They are equal at $p^* = \frac{3}{5}$, where both earn $\frac{6}{5}$.

ESS. Since $u(H,p^*) = u(D,p^*)$, every q satisfies $u(q,p^*) = u(p^*,p^*)$, so condition (i) never holds and we need (ii). By linearity in the first argument,

$$u(p^*,q) - u(q,q) = (p^* - q)\big[u(H,q) - u(D,q)\big] = (p^* - q)(3 - 5q) = 5\,(p^* - q)^2 > 0$$

for $q \neq p^*$, using $3 - 5q = 5(\frac{3}{5} - q)$. Hence $\boxed{p^* = \frac{3}{5}}$ is an ESS.

RUBRIC. 2 pts: neither pure strategy is an ESS. 2 pts: $p^* = \frac{3}{5}$. 1 pt: condition (i) fails, so (ii) must be checked. 3 pts: the identity $5(p^* - q)^2$ and conclusion.

(c) $\bar{u}(x) = x\,u(H,x) + (1-x)\,u(D,x)$, so $u(H,x) - \bar{u}(x) = (1-x)\big[u(H,x) - u(D,x)\big] = (1-x)(3-5x)$. Hence

$$\dot{x} = f(x) := x(1-x)(3-5x),$$

with rest points $x \in \{0, \frac{3}{5}, 1\}$. Now $f > 0$ on $(0, \frac{3}{5})$ and $f < 0$ on $(\frac{3}{5}, 1)$.

Let $x(0) \in (0, \frac{3}{5})$. Since f is a polynomial, solutions are unique, so $x(t)$ cannot reach the rest point $\frac{3}{5}$ in finite time and stays in $(0, \frac{3}{5})$; it is strictly increasing and bounded, so it converges to some $L \in (x(0), \frac{3}{5}]$. If $L < \frac{3}{5}$, then $f \geq c := \min_{[x(0),L]} f > 0$ along the trajectory and $x(t) \geq x(0) + ct \to \infty$, a contradiction. So $L = \frac{3}{5}$. The case $x(0) \in (\frac{3}{5}, 1)$ is symmetric, and $x(0) = \frac{3}{5}$ is a rest point. □

RUBRIC. 2 pts: derivation of $\dot{x}$. 1 pt: rest points. 3 pts: convergence argument (sign analysis alone: 1 pt; monotone convergence to a rest point: full).

Problem 4 *Stable matching* (20 points)

(a) Doctor-proposing deferred acceptance ("holds" are tentative):

Round	Proposals	Rejections	Held after round
1	$d_1 \to h_2$, $d_2 \to h_1$, $d_3 \to h_4$, $d_4 \to h_4$	h_4 rejects d_3	$h_1 : d_2$, $h_2 : d_1$, $h_4 : d_4$
2	$d_3 \to h_2$	h_2 rejects d_1	$h_1 : d_2$, $h_2 : d_3$, $h_4 : d_4$
3	$d_1 \to h_1$	h_1 rejects d_2	$h_1 : d_1$, $h_2 : d_3$, $h_4 : d_4$
4	$d_2 \to h_2$	h_2 rejects d_2	unchanged
5	$d_2 \to h_4$	h_4 rejects d_4	$h_1 : d_1$, $h_2 : d_3$, $h_4 : d_2$
6	$d_4 \to h_3$	none	$h_1 : d_1$, $h_2 : d_3$, $h_3 : d_4$, $h_4 : d_2$

Result: $\boxed{\mu_D = \{d_1h_1,\ d_2h_4,\ d_3h_2,\ d_4h_3\}}$.

RUBRIC. 4 pts: correct rounds (deduct 1 per incorrect rejection, max 4). 2 pts: final matching. Sequential (one proposal at a time) versions reach the same matching and receive full credit.

(b) *Termination and completeness.* Each doctor proposes to each hospital at most once, so the algorithm stops. Once a hospital receives a proposal it always holds one. If some doctor were rejected by all n hospitals, all n hospitals would be holding other doctors, but there are only $n-1$ other doctors. So everyone is matched.

Stability. Suppose (d, h) blocks the final matching μ: $h \succ_d \mu(d)$ and $d \succ_h \mu(h)$. Doctors propose in decreasing order of preference and d ends at $\mu(d)$, so d proposed to h before and was rejected at some point. A hospital rejects a doctor only in favor of one it strictly prefers, so the doctor held by h only improves over time; hence $\mu(h) \succ_h d$, a contradiction. □

RUBRIC. 2 pts: termination and everyone matched. 2 pts: d proposed to h earlier. 2 pts: monotonicity of hospitals' held offers and contradiction.

(c) Hospital-proposing: $h_1 \to d_3$, $h_2 \to d_4$, $h_3 \to d_1$, $h_4 \to d_2$. Each doctor receives exactly one proposal, so the algorithm ends after one round with

$$\boxed{\mu_H = \{d_1h_3,\ d_2h_4,\ d_3h_1,\ d_4h_2\}}.$$

Every stable μ satisfies $\mu_D(d) \succeq_d \mu(d) \succeq_d \mu_H(d)$. For d_2, $\mu_D(d_2) = \mu_H(d_2) = h_4$, so d_2 is matched to h_4 in *every* stable matching. For d_1, d_3, d_4 the partners in μ_D and μ_H differ, and both are stable, so these doctors do not have a unique stable partner. Answer: **only** d_2.

RUBRIC. 2 pts: μ_H. 2 pts: d_2 only, with the sandwich argument.

(d) In any pairing, D is paired with one of A, B, C.

- A with D (so B with C): C ranks A first, and A prefers C to D. So (A, C) blocks.

- B with D (so A with C): A ranks B first, and B prefers A to D. So (A, B) blocks.
- C with D (so A with B): B ranks C first, and C prefers B to D. So (B, C) blocks.

Every pairing is blocked, so no stable pairing exists. □

RUBRIC. 1 pt: reduction to who is paired with D. 1 pt per case.

Problem 5 *First-price auctions* (14 points)

(a) Suppose the rival bids $b(V) = \frac{2}{3}V$. Rival bids lie in $[0, \frac{2}{3}]$. A bid $\beta > \frac{2}{3}$ wins with probability 1 but pays more than $\frac{2}{3}$, which also wins with probability 1; so it is strictly worse than $\frac{2}{3}$. For $\beta \in [0, \frac{2}{3}]$ write $\beta = \frac{2}{3}w$ with $w \in [0, 1]$ ("bid as type w"). This bid wins iff $V < w$, with probability $F(w) = w^2$. The expected payoff is

$$\Pi(w) = \Big(v - \tfrac{2}{3}w\Big)w^2, \qquad \Pi'(w) = 2wv - 2w^2 = 2w(v - w).$$

So $\Pi' \geq 0$ on $[0, v]$ and $\Pi' \leq 0$ on $[v, 1]$, and $w = v$ (the bid $\frac{2}{3}v$) is a global maximizer. Hence b is a best response to itself: a symmetric BNE. □

RUBRIC. 1 pt: bids above $\frac{2}{3}$ dominated. 2 pts: winning probability w^2. 2 pts: derivative. 1 pt: global maximum via the sign of Π'. A first-order condition alone without the global argument: 4/6.

(b) First-price: the winner is the bidder with the higher value, $M = \max(V_1, V_2)$, with CDF x^4 and $E[M] = \frac{4}{5}$. Revenue $= \frac{2}{3}E[M] = \boxed{\frac{8}{15}}$. Second-price: revenue $= E[\min(V_1, V_2)] = \int_0^1 \Pr(\min > x)\, dx = \int_0^1 (1 - x^2)^2\, dx = 1 - \frac{2}{3} + \frac{1}{5} = \boxed{\frac{8}{15}}$. The revenues coincide, as revenue equivalence predicts (both auctions are efficient and a bidder with value 0 pays 0).

RUBRIC. 2 pts: first-price revenue. 2 pts: second-price revenue (1 pt for the correct integral with an arithmetic error).

(c) Let $c = \frac{2n-2}{2n-1} < 1$ and suppose the other $n - 1$ bidders bid cV_j. As in (a), bids above c are dominated by c, and a bid cw with $w \in [0, 1]$ wins iff all others have $V_j < w$, with probability $(w^2)^{n-1} = w^{2n-2}$. Then

$$\Pi(w) = (v - cw)\, w^{2n-2}, \qquad \Pi'(w) = w^{2n-3}\big[(2n-2)v - c(2n-1)w\big] = (2n-2)\, w^{2n-3}(v - w),$$

which is ≥ 0 for $w \leq v$ and ≤ 0 for $w \geq v$. So $w = v$ is optimal and $b_n(v) = cv$ is a symmetric BNE. □(For $n = 2$ this gives $\frac{2}{3}v$.)

RUBRIC. 1 pt: winning probability w^{2n-2}. 2 pts: derivative simplification. 1 pt: global optimality.

Problem 6 *Guessing two-thirds of the average* (14 points)

Write $p = \frac{2}{3}$.

(a) For fixed x_{-i}, u_i is a strictly concave quadratic in x_i maximized at $x_i = p\,m_{-i}$, which lies in $[0, 100]$. So $\boxed{BR_i(x_{-i}) = \frac{2}{3}m_{-i}}$. Since $m_{-i} \in [0, 100]$, the target $t = p\,m_{-i}$ lies in $[0, \frac{200}{3}]$. If $x_i > \frac{200}{3}$, then for every $t \le \frac{200}{3}$ we have $x_i - t > \frac{200}{3} - t \ge 0$, so $-(x_i - t)^2 < -(\frac{200}{3} - t)^2$: the strategy $\frac{200}{3}$ strictly dominates x_i.

RUBRIC. 2 pts: best response. 2 pts: dominance proof valid for all opponent profiles.

(b) Let S_k be the set surviving k rounds (the same for every player by symmetry). Claim: $S_k = [0, p^k \cdot 100]$. For $k = 0$ this is the strategy set. Suppose it holds for k and let $U = p^{k+1} \cdot 100$.

- Every $x_i > U$ is strictly dominated by U in the reduced game: when the others play in S_k, $m_{-i} \in [0, p^k \cdot 100]$ and $t = p\,m_{-i} \in [0, U]$, and the argument in (a) with $\frac{200}{3}$ replaced by U applies.
- Every $x_i \in [0, U]$ survives: if all others play $x_i/p \in [0, p^k \cdot 100] = S_k$, then $t = x_i$ and x_i earns 0, the maximum possible payoff. So no strategy does strictly better against this profile, and x_i is not strictly dominated.

Hence $S_{k+1} = [0, U]$, which completes the induction. The strategies that survive every round form the set $\bigcap_k [0, p^k \cdot 100] = \{0\}$, because $p^k \to 0$. □

RUBRIC. 1 pt: induction set-up. 2 pts: elimination above U. 2 pts: survival of all of $[0, U]$ (needed for "exactly"). 1 pt: conclusion.

(c) $(0, \ldots, 0)$ is an equilibrium: against $m_{-i} = 0$ the best response is 0. Conversely, in any equilibrium $x_i = p\,m_{-i}$ for all i. Let $M = \max_i x_i = x_{i^*}$. Then $M = p\,m_{-i^*} \le pM$, so $(1 - p)M \le 0$ and $M = 0$. Since all $x_i \ge 0$, every $x_i = 0$. □

RUBRIC. 1 pt: $(0, \ldots, 0)$ is an equilibrium. 3 pts: uniqueness via the maximum argument (or an equivalent argument, e.g. summing the best-response equations).